ACADEMIQ (AI-DRIVEN LMS)

MAJOR PROJECT-II

(20SSP65)

Submitted by

ANANDHACHITAN A

21MSS002

Under the Guidance of

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Associate Professor

Department of Software Systems

In partial fulfillment of the requirements for the award of the degree of

MASTER OF SCIENCE IN SOFTWARE SYSTEMS

(Five-Years Integrated Programme)

of Bharathiar University

DEPARTMENT OF SOFTWARE SYSTEMS

PSG COLLEGE OF ARTS & SCIENCE

An Autonomous College - Affiliated to Bharathiar University

Accredited with 'A^{++}'Grade by NAAC (4th Cycle)

College with Potential for Excellence

(Status Awarded by the UGC)

Star College Status Awarded by DBT - MST

An ISO 9001:2015 Certified Institution

Coimbatore - 641 014

April 2026

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CERTIFICATE

This is to certify that this project work entitled **"AI Recruiter"** is a bonafide record of work done by **Anandhachitan A (21MSS002)** in partial fulfillment of the requirements for the award of Degree of **Master of Science in Software Systems** (Five Years Integrated Course) of Bharathiar University.

_____ _____

Faculty Guide **Head of the Department**

Submitted for Viva-Voce Examination held on _____

_____ _____

Internal Examiner **External Examiner**

DECLARATION

I, **Anandhachitan A (21MSS002)**, hereby declare that this Project work entitled **"AI Recruiter"** is submitted to PSG College of Arts & Science (Autonomous), Coimbatore in partial fulfillment for the award of Master of Science in Software Systems, is a record of original work done by me under the supervision and guidance of **Dr. A. Subhashini M.Sc.,M.Phil.,MCA.,Ph.D.,SET.,NET.,** Associate Professor & Head of the Department, in Department of Software Systems, PSG College of Arts & Science, Coimbatore.

This report has not been submitted by me for the award of any other Degree/ Diploma/Associate ship/ Fellowship or any other similar degree to any other university

Place: Coimbatore **Anandhachitan A**

Date: **21MSS002**

ACKNOWLEDGEMENT

My venture stands imperfect without dedicating my gratitude to a few people who have contributed a lot towards the victorious completion for my project work.

I would like to thank **Thiru L. Gopalakrishnan, Managing Trustee, PSG & Sons Charities**, for providing me a prospect and surroundings that made the work possible.

I take this opportunity to express my deep sense of gratitude to **Dr. T. Kannaian M.Tech.,Ph.D.,** Secretary, PSG College of Arts & Science, Coimbatore for permitting and doing the needful towards the successful completion of this project.

I express my deep sense of gratitude and sincere thanks to our Principal **Dr. D. Brindha M.Sc.,M.Phil.,Ph.D.,MA (Yoga).,** for her valuable advice and concern for students. I am very thankful to **Dr. M Umarani M.B.A.,M.Phil.,Ph.D.,** Vice Principal ,for their support towards my project.

I sincerely thank **Dr. K.V. Rukmani MCA.,M.E.,Ph.D.,** Associate Professor and Head of the Department, Department of Software Systems for her wholehearted help to complete this project successfully by giving valuable suggestions.

I convey my heartiest and passionate sense of thankfulness to my project guide

Dr. A. Subhashini M.Sc.,M.Phil.,MCA.,Ph.D., Assistant Professor, Department of Software Systems, for his timely suggestion which enabled me in completing the project successfully.

I express my sincere gratitude to **Mr. Sajish Kumar Athra Pulikkal Mohanan,Assistant Manager - Web Development, GIBS - Digital - eCommerce Development, Avantor, Inc.**, for providing me with valuable guidance, continuous support, and expert insights throughout the development of this project.

This note of acknowledgment will be incomplete without paying my heartful devotion to my parents, my friends, and other people, for their blessings, encouragement, financial support, and the patience, without which it would have been impossible for me to complete the job.

Anandhachitan A

21MSS002

INTERNSHIP OFFER LETTER

avantor™

VWR Lab Products Private Limited
Rathinam Techno Park, Pollachi Main Road,
Eachanari Post, Coimbatore - 641021
Regd Office
Unit No. 10, Ground Floor, Empire Infantry Building,
Plot No. 29, Infantry Road,
Bangalore 560001

September 19, 2025

Perumal Palaniappan
710, Vinayagar Kovil Street,
Lawly Road,
Coimbatore - 641003

Mobile: 8838375900
Email: perumal.palaniappan04@gmail.com
College/University: PSG College Of Arts & Science

Subject: Internship opportunity at VWR Lab Products Private Limited

Dear Perumal Palaniappan,

We write to you for and on behalf of VWR Lab Products Private Limited having its registered office at Unit No. 10, Ground Floor, Empire Infantry Building, Plot No. 29 (old Plot no. 10), Infantry Road, Bangalore-560001, Karnataka, India (hereinafter referred to as "**Company**"), with reference to your application for an internship at the Company.

We are pleased to offer you an internship opportunity with the Digital Engineering department based at Coimbatore. You will be reporting to Sajish Kumar Athra Pulikkal Mohanan (**"Reporting Manager"**). The Reporting Manager in consultation with you will develop a program description and expectations designed to achieve the objectives of the internship. The internship will provide you with valuable learning experience and is directly related to the subject matter of your studies as a full-time student at your College/University. Company's goal in developing this internship is to help you gain first-hand experience and for you to observe how your course of study is applied in a real-world setting and improve your skills in those areas.

Scope of Internship: Student Internship with the Digital Engineering Team.

Duration of the Internship: Six (6) months, i.e., from September 19, 2025 to March 19, 2026.
The duration of the internship may be extended at the sole discretion of the Company.

Stipend: During the duration of the internship you shall be paid an amount of INR 20,000/- per month (hereinafter referred to as "Stipend") for the expenses that you may incur during the duration of the internship. You shall not be entitled to receive any other payment from the Company during the internship, unless and until confirmed in writing by the Company. All taxes payable on the Stipend shall be borne by you and the Company shall have no liabilities in this respect. The Stipend so payable by the Company to you as provided above shall be subject to applicable income-tax deductions at source, if any. Payment of the stipend shall not be construed as creating any employment relationship between you and the Company.

You hereby confirm and acknowledge you are participating in the internship and that the internship will be conducted without entitlement to a paid job and/or any other form of employment with the Company at the conclusion of the internship.

The location of your internship will be the Coimbatore office of the Company. During the internship period, you will be required to attend office regularly on all working days. In case of any inability to attend office or if you need to take leave, you must inform your Reporting Manager in advance.

Perumal.P

INTERNSHIP COMPLETION LETTER

Rndsoft
An ISO Certified Company

September 03, 2024

TO WHOMSOEVER IT MAY CONCERN

This is to state that **Mr. PERUMAL.P,** Register No –**21MSS034**, 4th Year **MSC SOFTWARE SYSTEMS** Student from **"PSG COLLEGE OF ARTS AND SCIENCE"** has successfully completed his **SOFTWARE** Internship training from **JUNE 03, 2024** to **SEPTEMBER 03, 2024** in our company.

We wish him every success in life.

RND SOFTECH PRIVATE LIMITED Rndsoft COIMBATORE

For RND Softech Private Limited,

Asst. Manager – Talent Acquisition

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SYNOPSIS

AI Recruiter is an AI-powered recruitment platform designed to modernize and streamline the hiring process through intelligent automation and a user-friendly interface. The system integrates modules such as interview scheduling, candidate evaluation, recruiter dashboards, and automated reporting into a single platform. It uses modern frontend technologies like Next.js and Tailwind CSS to deliver a responsive, fast, and seamless experience tailored specifically for recruiters.

The platform incorporates secure authentication through Supabase Google Auth, ensuring frictionless login and secure session management. Recruiters are provided with a dedicated dashboard that centralizes interview creation, candidate tracking, and evaluation reports, enabling structured workflows and controlled access. The backend, powered by Node.js, Express.js, and Supabase's managed PostgreSQL, efficiently handles candidate data while maintaining scalability, security, and smooth API communication across the system.

A key highlight of AI Recruiter is its integration with advanced AI services such as GPT-based models for dynamic interview question generation and Vapi for voice agent functionality and speech-to-text transcription. These intelligent features enable automated interviews, structured evaluation reports, and personalized candidate feedback. By combining a powerful technical stack with Supabase authentication and smart AI capabilities, AI Recruiter delivers a next-generation recruitment solution that is efficient, interactive, and adaptable to the evolving needs of modern hiring practices.

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CHAPTER 1

INTRODUCTION

1.1 COMPANY PROFILE

Avantor is a global life-science tools company that supplies mission-critical products, materials, and services to customers in biopharma, healthcare, education and government, and advanced technologies, operating through a worldwide manufacturing, distribution, and digital network. The VWR brand and the vwr.com channel form a core part of Avantor's go-to-market model, serving customers in over 180 countries and shipping to hundreds of thousands of locations across the globe. These operations are supported by a broad portfolio and a dedicated e-commerce ecosystem designed to streamline laboratory procurement and services.

In India, VWR functions as Avantor's channel for laboratory products and services, offering local commercial support and access to the global catalogue through region-specific portals and contact hubs. This integrated presence connects Indian customers with Avantor's worldwide supply chain and service standards, ensuring consistent access to laboratory consumables, equipment, chemicals, and related solutions.

The Coimbatore site is publicly referenced as a Global Business Center that supports VWR/Avantor's back-end and shared-services operations, located at Rathinam Techno Park, Eachanari, on Pollachi Main Road. Listings describing the campus identify its role in corporate services and provide local switchboard details, reflecting the site's position within Avantor's broader channel and shared-services strategy. These sources place the facility within the Eachanari/Coimbatore Industrial Estate area and highlight its growth as part of VWR's India footprint.

1.1.1 CONTACT

Company: VWR Lab Products Private Limited (VWR, part of Avantor) Coimbatore

Address: Rathinam Techno Park, Eachanari (PO), Pollachi Main Road, Coimbatore, Tamil Nadu 641021, India

Phone: +91 422 434 3300

1.2 PROJECT OVERVIEW

AI Recruiter is an AI-driven recruitment platform developed to transform hiring workflows through automation, intelligent assistance, and a modern recruiter-centric interface. The platform enables seamless management of interviews, candidate evaluations, feedback reports, and recruiter dashboards within a single system. Built with Next.js and Tailwind CSS on the frontend and powered by Node.js, Express.js, and Supabase on the backend, AI Recruiter ensures a smooth, scalable, and secure environment for digital recruitment. It supports authentication through Supabase Google Auth, offering a secure and frictionless login experience while enforcing structured access for recruiters.

A highlight of AI Recruiter is its integration of advanced AI services such as GPT-based models for dynamic interview question generation and Vapi for voice agent capabilities and speech-to-text transcription. These intelligent features enable automated interviews, structured evaluation reports, and personalized candidate feedback. By combining a robust technical stack with Supabase authentication and smart AI capabilities, AI Recruiter delivers an innovative recruitment solution that enhances productivity, reduces manual workload, and supports modern hiring needs with efficiency and intelligence.

1.3 MODULE DESCRIPTION

- **Authentication & User Access Module**

This module handles secure login through traditional credentials as well as OAuth providers like Google and GitHub. Users select their role during registration, and the system automatically assigns permissions and redirects them to their respective dashboards. It ensures proper session handling, protected routes, and secure access control for all roles.

1.3.1 STUDENT MODULES

- **Student Dashboard Module**

The student dashboard displays personalized learning insights such as total courses, generated notes, flashcards, and quiz status. It provides quick access to AI tools and recently viewed courses, helping students resume their learning seamlessly. A visual activity graph highlights their weekly learning performance.

- **Explore Courses & Learning Module**

Students can browse and enroll in courses categorized by subject and difficulty. Each course provides structured modules, downloadable resources, and lesson-wise content. The module ensures smooth navigation through the entire learning journey.

- **AI Notes Generator Module**

Students can generate instant notes on any topic using Cohere Command-A by selecting difficulty and content preferences. The generated notes are neatly structured for learning and revision. Recent generations are stored for quick future reference.

- **AI Flashcards Generator Module**

This module allows students to create topic-based flashcards by selecting difficulty levels and card count. The AI generates concise, easy-to-remember flashcards to support faster revision. Students can save and reuse flashcards anytime.

- **Quiz Module**

Students can attempt quizzes assigned by instructors, categorized into Pending, Completed, and New. Each quiz displays the number of questions and instructor details before starting. After submission, students can view scores and track their quiz performance history.

1.3.2 INSTRUCTOR MODULES

- **Instructor Dashboard Module**

The instructor dashboard highlights metrics like total courses created, enrolled students, and quizzes assigned. A graphical overview shows how students interact with the instructor's content. Shortcut buttons allow quick creation of courses and quizzes.

- **Course Creation & Management Module**

Instructors can create structured courses with banners, objectives, and multi-module content. They can also specify special features like certificates or downloadable PDFs. All created courses can be edited, published, or managed from the "My Courses" section.

- **Quiz Creation & Management Module**

This module enables instructors to create quizzes with various question types such as MCQs and checkboxes. Instructors can set titles, instructions, and publish quizzes to students. They can also track quiz status (Pending/Accepted) and edit them when needed.

1.3.3 SUPERADMIN MODULES

- **SuperAdmin Dashboard Module**

The SuperAdmin dashboard provides platform-wide analytics such as total users, courses, quizzes, and recent registrations. Graphs visualize user growth and platform activity in real time. It also includes quick controls for managing users, courses, and quizzes.

- **User Management Module**

SuperAdmins can view all platform users, update their roles, or remove inactive accounts. This module ensures only authorized individuals have access to sensitive or creation-based features. It maintains the overall stability and integrity of the platform.

- **Course & Quiz Oversight Module**

SuperAdmins can review all courses and quizzes, ensuring quality and compliance with platform guidelines. They can adjust visibility, monitor instructor activity, and maintain content standards. This module ensures smooth academic workflow across the system.

1.3.3 GLOBAL MODULE

- **Settings & Account Management Module**

Users can update personal information, modify passwords, and switch between light and dark themes. The settings area centralizes all customization options for better user control. It also contains a "Danger Zone" that allows users to permanently delete their accounts.

CHAPTER 2

SYSTEM ANALYSIS

2.1 EXISTING SYSTEM

The current recruitment processes rely heavily on manual scheduling, interviewer availability, and subjective evaluation, which increases time consumption and reduces efficiency. Many hiring platforms lack integrated AI features, forcing recruiters to depend on external tools for conducting interviews, transcribing responses, and generating evaluation reports. Authentication systems in traditional recruitment applications are often basic and do not provide seamless integration with modern authentication providers. Monitoring and analytics tools are limited, restricting visibility into candidate performance and overall recruitment activity. Recruiters usually do not have access to automated assistants for conducting interviews or generating structured feedback, resulting in repetitive tasks and inconsistent evaluation standards.

2.2 PROPOSED SYSTEM

AI Recruiter introduces an AI-powered recruitment platform that automates key hiring processes such as interview scheduling, voice-enabled interviews, and structured evaluation reporting using GPT models and Vapi for voice agent and speech-to-text transcription. The system integrates secure and modern authentication through Supabase Google Auth, ensuring seamless access and enhanced security. A recruiter-centric dashboard is implemented, offering centralized control over interview creation, candidate tracking, and feedback management to streamline workflow. Recruiters can efficiently generate dynamic interview questions, conduct automated voice interviews, and receive AI-driven evaluation reports, reducing manual workload and improving consistency.

The platform also provides advanced analytics and monitoring tools, enabling recruiters to gain insights into candidate performance and overall hiring activity. By combining structured workflows, intelligent automation, and modern web technologies, AI Recruiter enhances productivity, reduces bias, and delivers a next-generation recruitment solution tailored to the evolving needs of digital hiring.

CHAPTER 3

SYSTEM CONFIGURATION

3.1 HARDWARE SPECIFICATION

COMPONENTS	REQUIREMENTS
• Processor	: Intel Core i3 (10th Gen) / AMD Ryzen 3
• Hard Disk	: 256 GB SSD
• RAM	: 8 GB
• System Type	: 64-bit operating system, x64-based processor
• Keyboard	: Standard 104-key keyboard
• Mouse	: Optical Mouse

3.2 SOFTWARE SPECIFICATION

COMPONENTS	REQUIREMENTS
• Operating System	: Windows 10 / Windows 11
• Frontend Development Tools	: Next.js, Tailwind CSS
• Backend Development Tools	: Node.js, Express.js
• Database & Authentication	: Supabase (PostgreSQL + Google Auth)
• AI Integration	: GPT-4o, Vapi
• Version Control	: Git & GitHub
• Execution Environment	: Visual Studio Code
• Browser	: Google Chrome / Microsoft Edge

CHAPTER 4

SOFTWARE DESCRIPTION

4.1 Node.js (Backend Runtime Environment)

Node.js is a high-performance JavaScript runtime built on Chrome's V8 engine, enabling fast and non-blocking backend server execution. It is widely used to develop REST APIs, handle asynchronous processes, and support scalable server-side logic. Node.js integrates seamlessly with npm (Node Package Manager), allowing developers to install thousands of libraries for authentication, routing, database connectivity, and more. Its event-driven architecture ensures efficient handling of multiple client requests.

4.2 Express.js (Web Framework)

Express.js is a lightweight and flexible Node.js framework used to build backend APIs and routing structures. It simplifies server creation, middleware management, and request handling, making development faster and more organized. In AI Recruiter, Express.js powers secure routing, authentication middleware, and integration with Supabase and Vapi services. Its modular structure enables maintainable and clean backend code, ensuring recruiters can reliably access dashboards, candidate data, and AI-driven interview features.

4.3 Next.js (Frontend Library)

Next.js is a powerful React-based framework that enables server-side rendering (SSR), static site generation (SSG), and optimized routing for modern web applications. It improves performance, SEO, and scalability by pre-rendering pages and delivering fast, interactive experiences. In AI Recruiter, Next.js is used to build recruiter dashboards, interview creation forms, and evaluation report pages with seamless navigation and responsive layouts. Its integration with Supabase APIs and AI services ensures recruiters experience smooth workflows and real-time updates during the hiring process.

4.4 Tailwind CSS (Styling Framework)

Tailwind CSS is a utility-first CSS framework that enables rapid and responsive UI development. It eliminates the need for writing large CSS files by providing pre-built classes that can be applied directly within JSX components. Tailwind helps create clean, modern, and mobile-friendly designs throughout the recruiter interface. It also supports

full customization through configuration files for colors, spacing, typography, and themes, ensuring AI Recruiter maintains a consistent and professional look across devices.

4.5 Supabase (Database & Authentication Platform)

Supabase is an open-source backend-as-a-service built on PostgreSQL, providing managed database storage, authentication, and real-time APIs. In AI Recruiter, Supabase handles secure recruiter authentication via Google Auth and manages candidate data, interview logs, and evaluation reports. Its integration with Express.js ensures smooth API communication, while its scalable infrastructure supports reliable data storage and retrieval. Supabase simplifies backend operations by combining database management and authentication into a single platform.

4.6 GPT-4o & Vapi (AI Integration)

GPT-4o powers intelligent recruitment tools such as dynamic interview question generation, candidate evaluation, and automated feedback reporting. It delivers accurate, context-aware outputs based on recruiter preferences and candidate responses. Vapi provides voice agent functionality and speech-to-text transcription, enabling AI Recruiter to conduct automated voice interviews and convert candidate answers into structured text for analysis. Together, GPT-4o and Vapi automate critical recruitment tasks, reduce manual workload, and enhance the overall efficiency of the hiring process.

CHAPTER 5

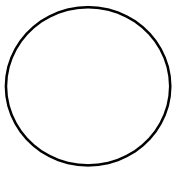
SYSTEM DESIGN

5.1 DATA FLOW DIAGRAM

A data flow diagram (DFD) is a graphical representation of the "flow" of data through an information system. DFD can also be used for the visualization of data processing. The purposes of DFD are as follows:

- To show the scope and boundaries of a system.
- To show that the whole system has been considered.
- May be used as a communication tool between a system analyst and any person who plays a part in the system.
- To act as the starting point for redesigning a system.

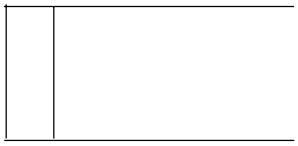
The representation used to frame a data flow diagram is:



- The circle represents the process. It shows the pair of processes that transform input to output.

- The arrow represents the graphical flow into or out of the process.

- The terminator represents external entities with systemcommunication.



- The store is used to model a collection of data packets at rest.

Level 0:

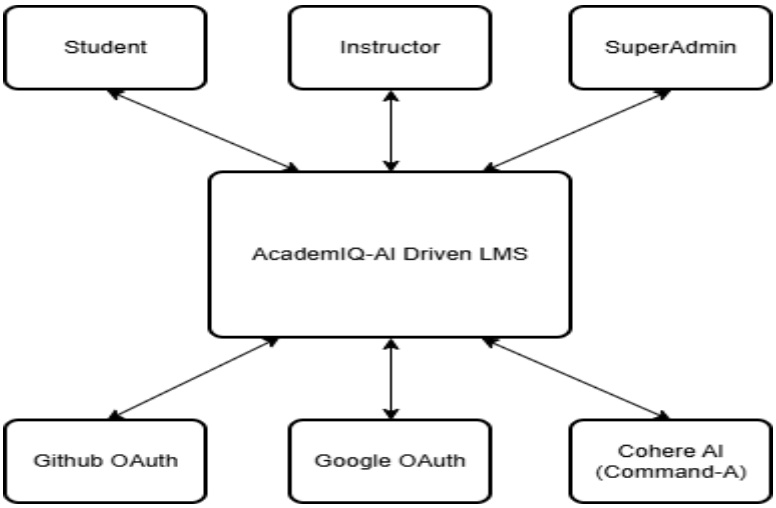


Figure 5.1.1 Level 0 DFD

Level 1:

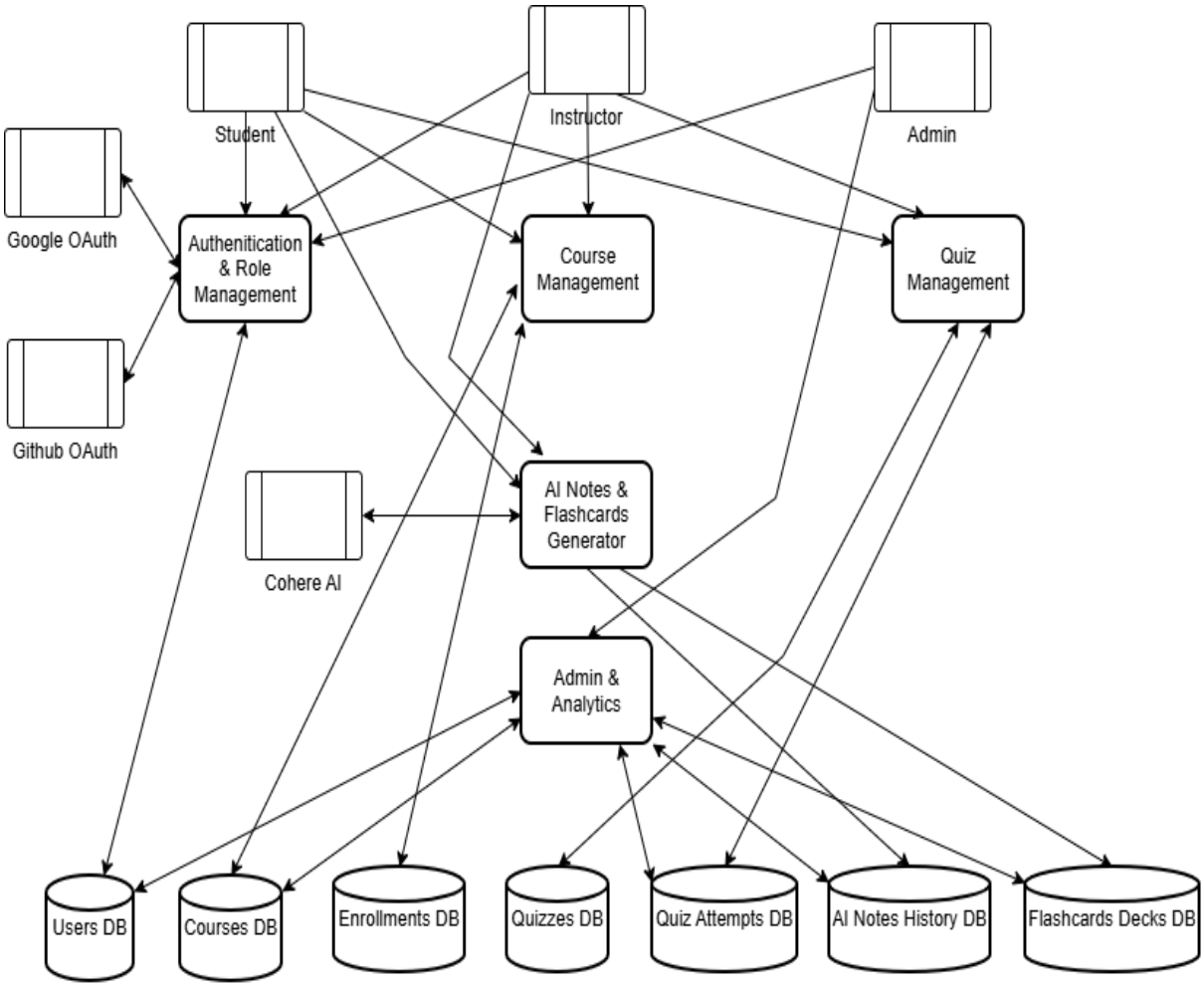


Figure 5.1.2 Level 1 DFD

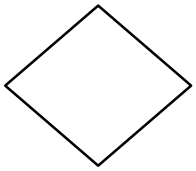
5.2 ENTITY RELATIONSHIP DIAGRAM

An Entity Relationship Diagram (ERD) is a specialized graphic that illustrates the relationship between entities in a database. It is a data modeling technique that graphically illustrates an information system's entities and the relationships between those entities.

An ERD is a conceptual and representational model of data used to represent the entity framework infrastructure.

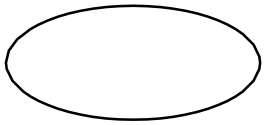
The elements of an ERD are:

- Entities
- Relationship
- Attributes



- The diamond is used to represent relationships

- The box is used to represent the entities



- The oval is used to represent the attributes

- The line is used to represent the flow

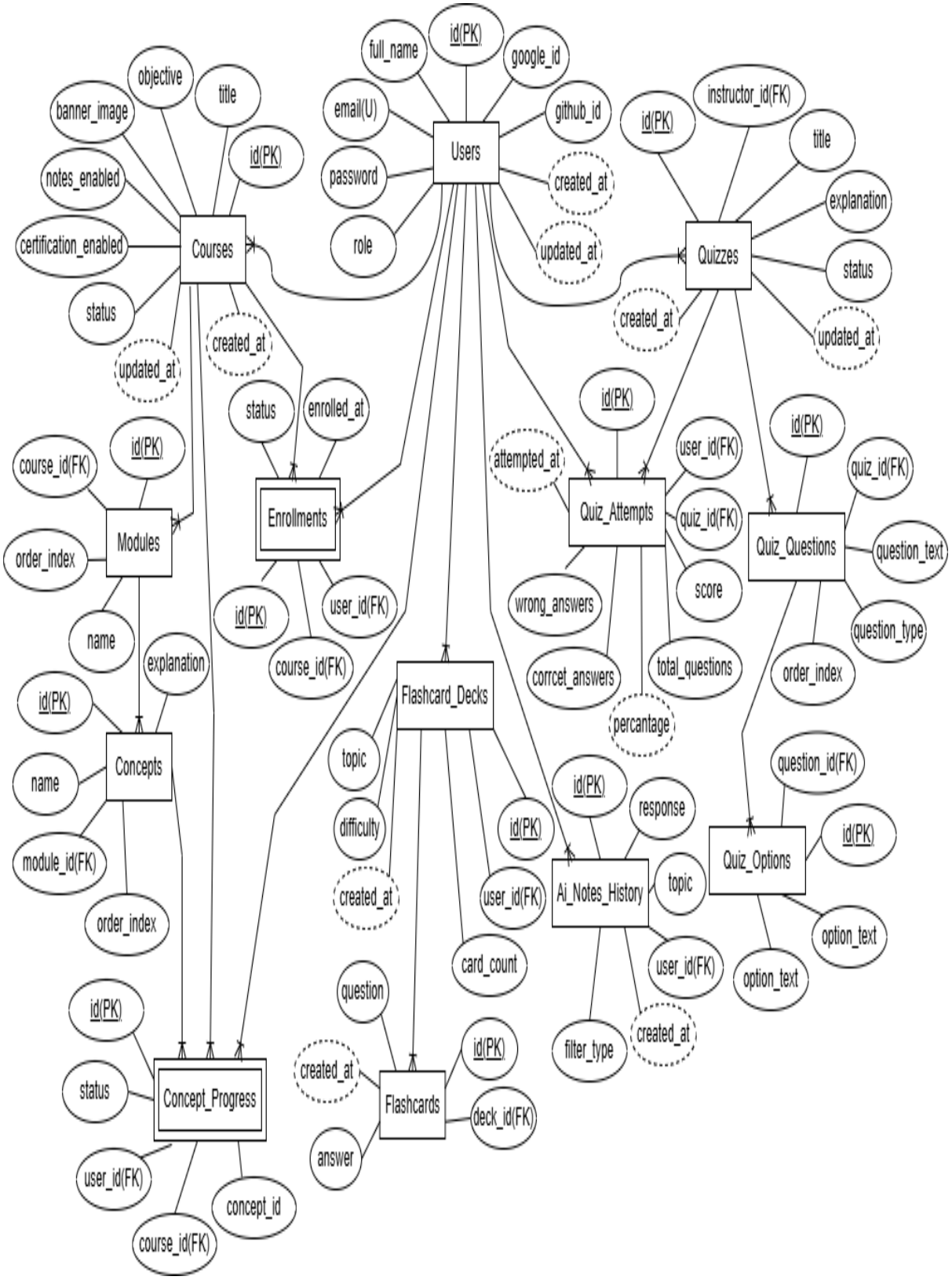


Figure 5.2.1 Entity Relationship Diagram

5.3 INPUT DESIGN

The input design of AI Recruiter focuses on creating a structured, user-friendly, and secure method for capturing all necessary information from recruiters during the hiring process. It ensures that each form such as login, interview creation, candidate details entry, and question generation collects accurate data through validated fields and guided input patterns. The system uses dropdowns, text fields, date/time pickers, and file upload controls to simplify data entry while minimizing user errors. Proper client-side and server-side validation ensures that inputs such as candidate information, interview parameters, and authentication credentials are processed reliably. Overall, the input design guarantees a smooth and accurate data flow into the system, supporting both usability and data integrity for recruiters.

5.4 OUTPUT DESIGN

The output design of AI Recruiter delivers clear, interactive, and visually organized results to recruiters through dashboards, analytics charts, interview reports, and AI-generated evaluations. Information is presented in a structured layout using cards, tables, and formatted content to help recruiters easily understand candidate performance and interview outcomes. Recruiters view interview schedules, candidate responses, evaluation scores, and AI-generated feedback summaries, all displayed in real time. The system also provides structured reports that can be exported or shared, ensuring recruiters have actionable insights for decision-making. This output design ensures that system responses are meaningful, accessible, and reliable, enabling recruiters to conduct efficient and data-driven hiring processes.

CHAPTER 6

SYSTEM IMPLEMENTATION AND TESTING

6.1 SYSTEM IMPLEMENTATION

The implementation of AI Recruiter involved building a complete AI-powered recruitment platform using a structured and modular development approach. The frontend was developed using Next.js and Tailwind CSS, enabling server-side rendering, dynamic page generation, smooth navigation, and consistent UI components across recruiter dashboards, interview creation forms, candidate tracking pages, and analytics screens. Every recruiter interaction from login to interview scheduling was mapped into responsive components to ensure seamless experience on different devices.

On the backend, Node.js and Express.js were used to build secure REST APIs responsible for authentication, interview management, candidate data handling, evaluation reporting, and AI services. Supabase was chosen as the database and authentication provider because of its managed PostgreSQL structure and built-in support for Google Auth. Supabase efficiently manages entities such as recruiters, candidates, interviews, questions, and evaluation reports. Proper relational mapping and optimized queries ensured fast data retrieval, even with multiple interconnected tables.

AI integration was implemented using OpenRouter with GPT-4 mini for dynamic interview question generation, candidate evaluation, and automated feedback reporting. In addition, Vapi was integrated to provide voice agent functionality and speech-to-text transcription, enabling the system to conduct automated voice interviews and convert candidate responses into structured text for analysis. Supabase authentication with Google login was added to simplify recruiter onboarding and enhance security. Each recruiter module was implemented with route protection, ensuring that only authorized users could access sensitive features

6.2 SOFTWARE TESTING

Testing is a process, which reveals errors in the program. It is the major quality measure employed during software development. During testing, the program is executed with a set of conditions known as test cases, and the output is evaluated to determine whether the program is performing as expected.

The process of testing the functionality and correctness of software by running it is called software testing. It is the process of executing a program with the intent of finding an error. A good test case has a high probability of finding an undiscovered error. Software testing is usually performed for two main reasons:

- Defect detection
- Reliability estimation.

6.2.1 Functional Testing

Functional testing was performed to verify that every feature works according to the system requirements. All major modules authentication, interview scheduling, candidate data entry, question generation, voice interview handling, and evaluation reporting were executed and validated. Each screen and API was tested to ensure accurate input processing and correct output generation. This guaranteed that recruiter actions such as creating interviews, viewing reports, and managing candidate details functioned smoothly without errors.

6.2.2 Integration Testing

Integration testing ensured proper communication between the frontend, backend, database, and AI services. Next.js components were tested against Express APIs to confirm that data was transmitted and received correctly. Supabase interactions, such as storing candidate details, retrieving interview logs, and updating evaluation reports, were validated thoroughly. This testing confirmed that combined modules worked together as a unified system.

6.2.3 User Interface (UI) Testing

User interface testing checked the consistency, responsiveness, and clarity of all screens such as recruiter dashboards, interview creation forms, and evaluation report pages. The UI was validated across different screen sizes to ensure that layouts, buttons, forms, and navigation elements displayed properly. This testing confirmed that the design was intuitive and user-friendly for recruiters.

6.2.4 Security Testing

Security testing focused on access protection, token validation, and route restrictions. Routes were tested to ensure unauthorized users could not access recruiter dashboards or sensitive data. Input validation was checked to prevent invalid data submission. Supabase authentication with Google login was tested to confirm secure handling of recruiter sessions..

6.2.5 User Acceptance Testing (UAT)

User acceptance testing involved navigating through the entire system like a real recruiter to validate usability and end-to-end functionality. All workflows login, creating interviews, generating questions, conducting voice interviews, and viewing evaluation reports were executed in real conditions. This testing confirmed that the system meets recruiter expectations and provides a smooth, reliable experience.

CHAPTER 7

CONCLUSION & FUTURE RECOMMENDATION

7.1 CONCLUSION

AI Recruiter demonstrates how modern web development and artificial intelligence can come together to create a practical and effective recruitment platform. The system integrates a responsive user interface with reliable server-side services and a well-structured relational database so that every feature works smoothly from recruiter login and authentication to interview scheduling, candidate evaluation, and analytics. The inclusion of **OpenRouter GPT-4 mini** for generating interview questions and structured feedback, combined with **Vapi's voice agent and speech-to-text transcription**, reduces manual effort for recruiters and provides faster access to consistent evaluation reports. Supabase authentication ensures secure access, while the recruiter-centric dashboard delivers the right tools and views to simplify daily hiring operations.

Through careful implementation and systematic testing, the platform has proven to be stable, secure, and ready for real usage. Functional checks verified that every workflow behaves as intended. Integration checks validated data flow across the client, the server, the database, and external AI services. Interface checks confirmed that pages are clear and consistent across devices. The overall outcome is a dependable recruitment system that improves hiring productivity, supports structured candidate evaluation, and gives recruiters clear visibility and control. AI Recruiter provides a solid foundation for future extensions such as richer analytics, multilingual interview support, ATS (Applicant Tracking System) integration, and deeper automation of recruitment workflows.

7.2 FUTURE RECOMMENDATION

- **Advanced Candidate Analytics and Predictive Insights**

Future versions can include deeper analytics such as prediction of candidate success, identification of skill gaps, and automated insights for recruiters. This would help in making data-driven hiring decisions and improve the accuracy of candidate selection.

- **Real-Time Virtual Interview Rooms**

A live interview module can be added where recruiters conduct real-time sessions using video streaming and integrated voice agents. Features such as attendance tracking, live chat, and interactive evaluation notes would further strengthen the recruitment environment.

- **Adaptive Question Generation**

The system can be enhanced with adaptive AI algorithms that analyze candidate responses and automatically adjust the difficulty or type of interview questions. This would help personalize the interview experience and provide recruiters with more accurate assessments.

- **Enhanced Candidate Identity Verification**

Future versions can include advanced security features such as mandatory ID proof submission and automated face matching against passport-size photos. This would help recruiters verify candidate authenticity, reduce impersonation risks, and ensure that only genuine applicants participate in interviews.

- **AI-Usage Detection During Interviews**

The system can be expanded with mechanisms to identify whether candidates are relying on external AI tools to answer questions. By analyzing response patterns, timing, and consistency, recruiters can gain insights into the originality of candidate answers and ensure fair evaluation.

- **Eye-Tracking for Focus Validation**

Future improvements can include lightweight eye-tracking technology that monitors whether candidates maintain focus on the camera for 3–4 seconds during interviews. This feature would help detect distractions such as switching to other tabs or using mobile devices, ensuring that candidates remain attentive and engaged throughout the session.

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APPENDIX

SCREENSHOTS

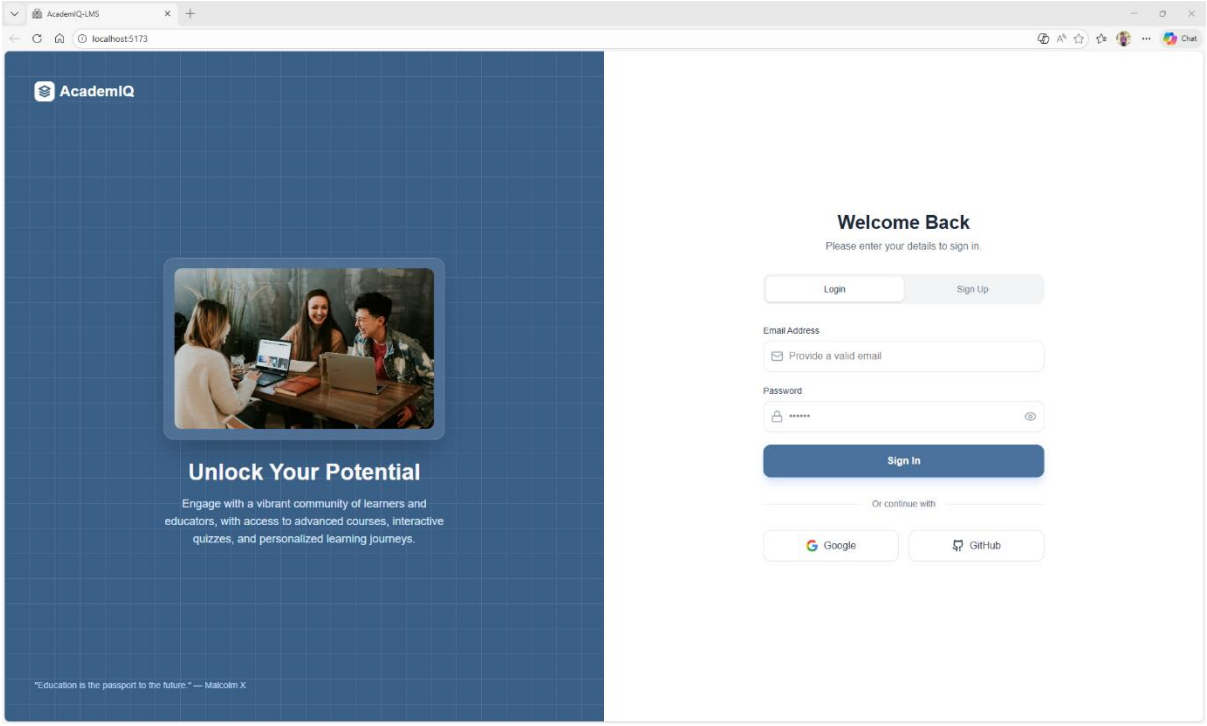

Figure. Login Page

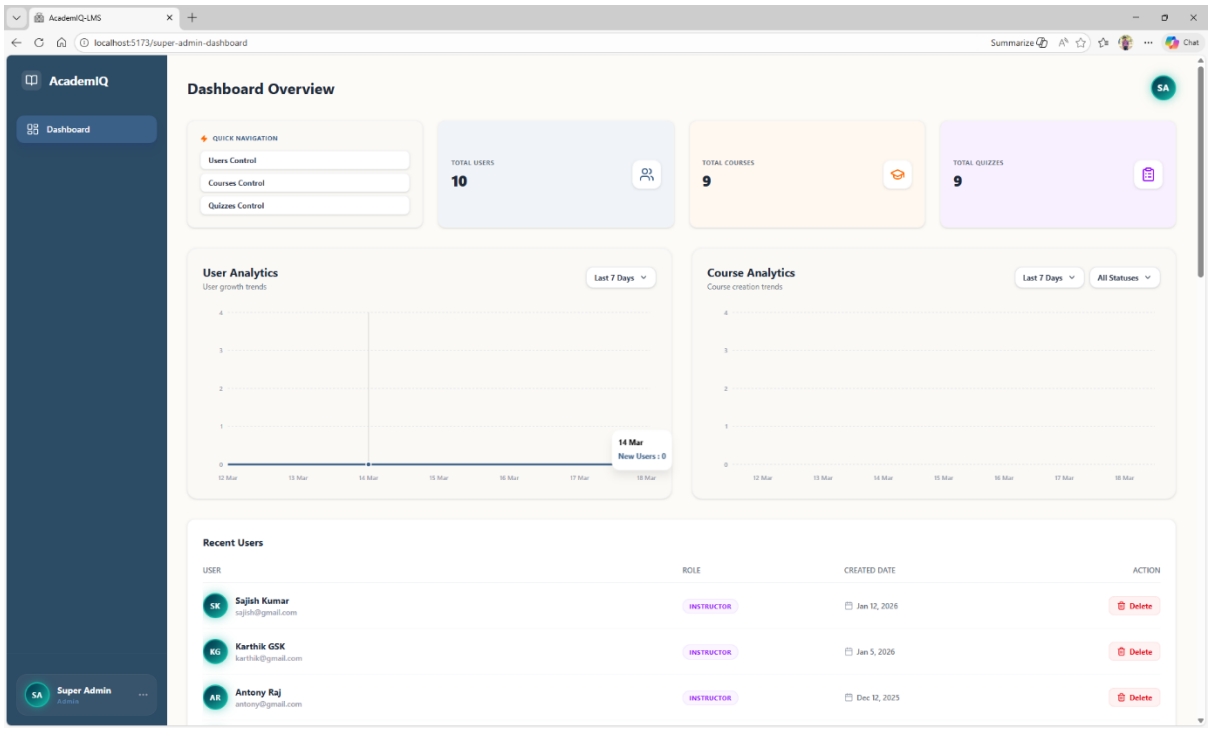

Figure. Admin Control Page

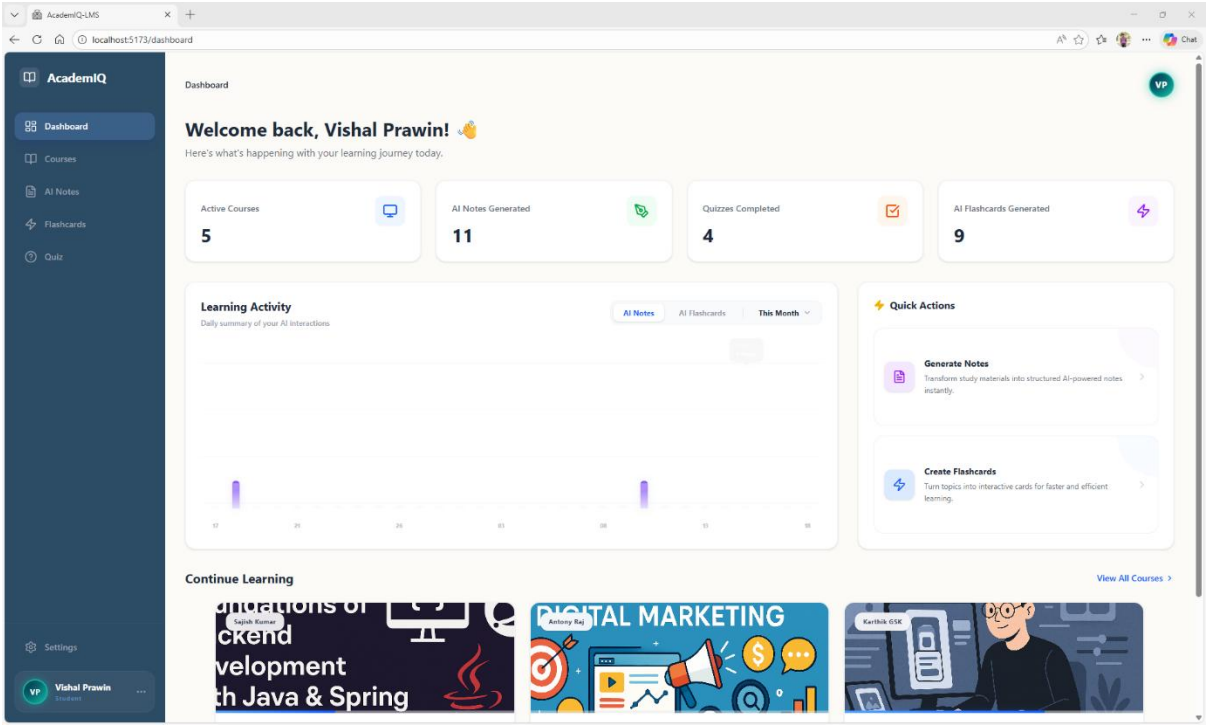

Figure. Dashboard Page

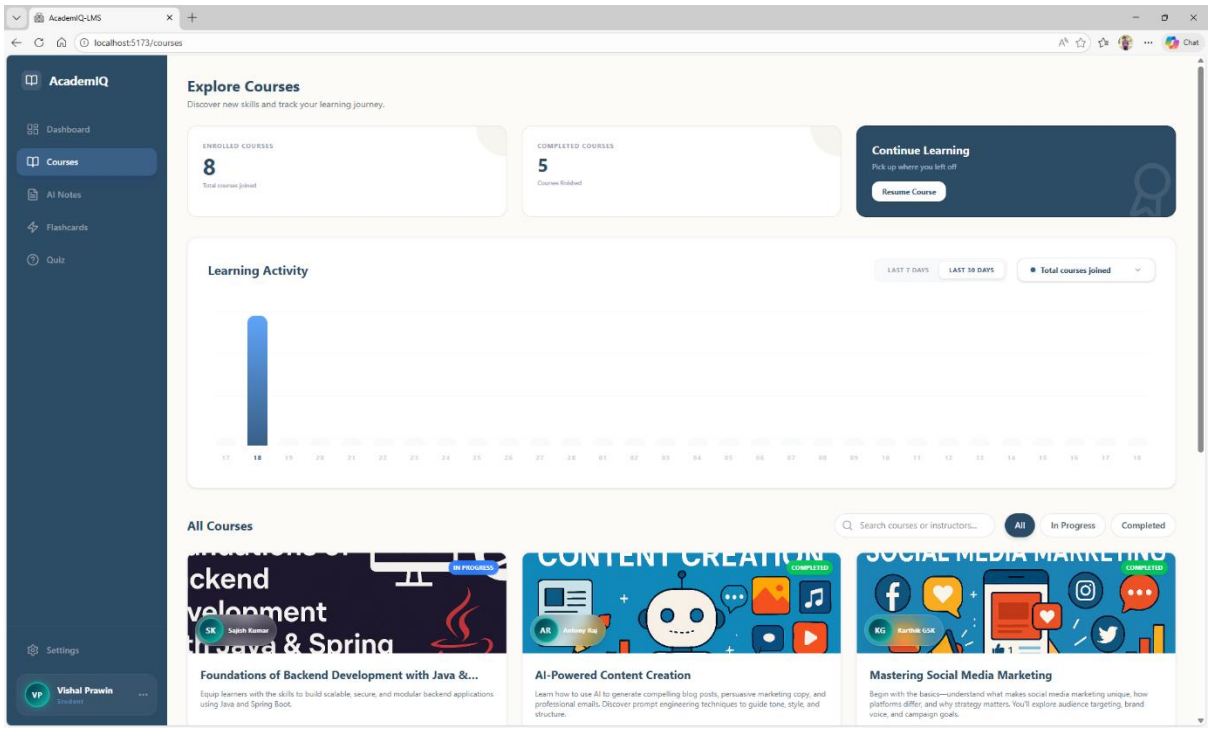

Figure. Courses Page

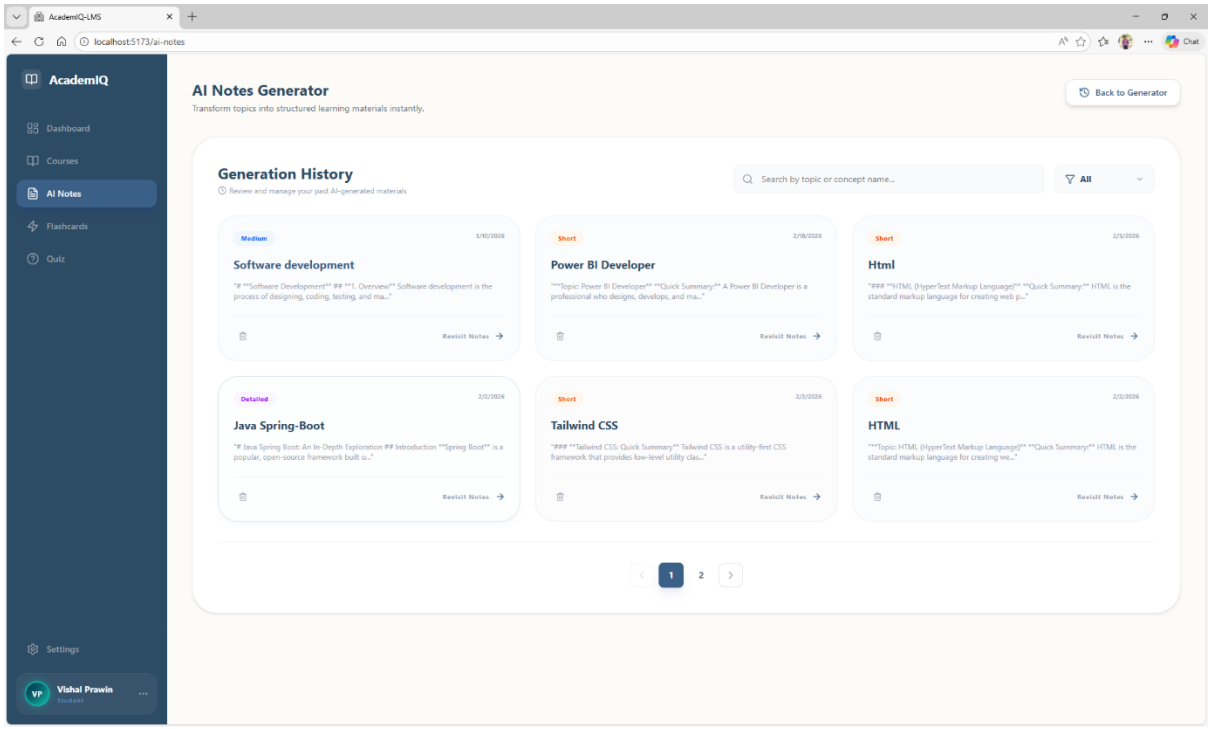

Figure. AI Notes Generation Page

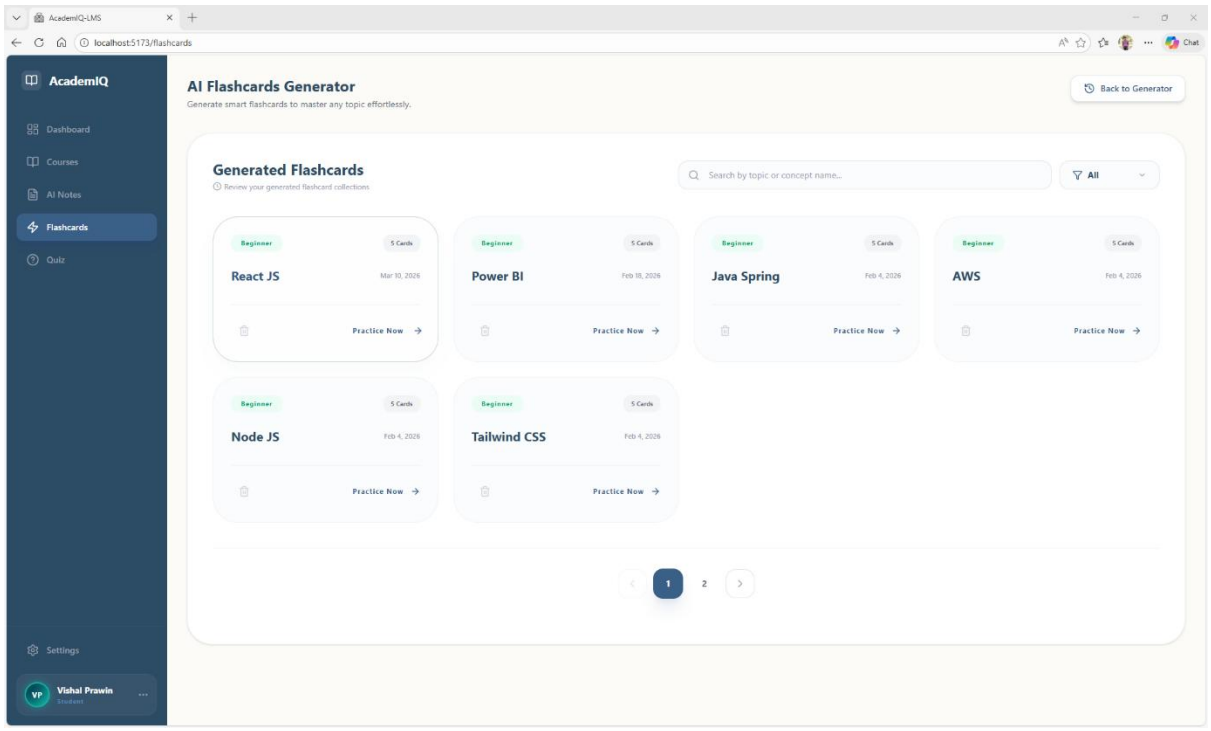

Figure. AI Flashcards Page

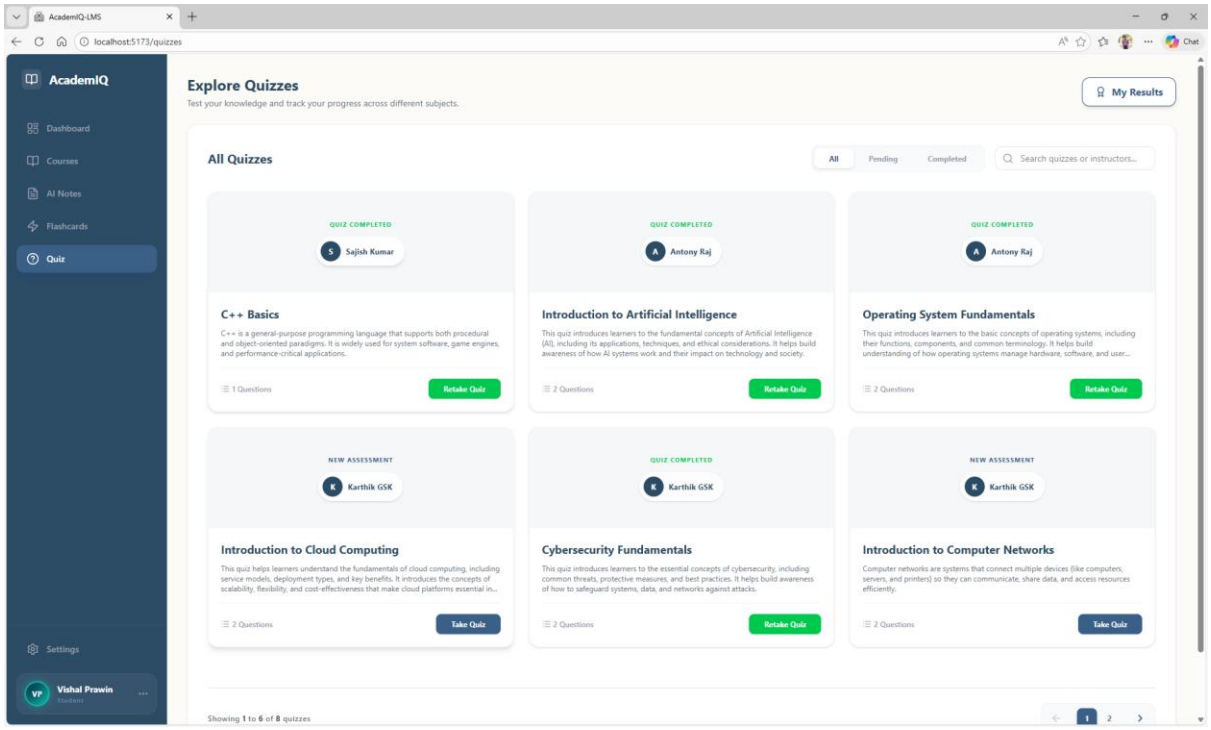

Figure. Quizzes Page

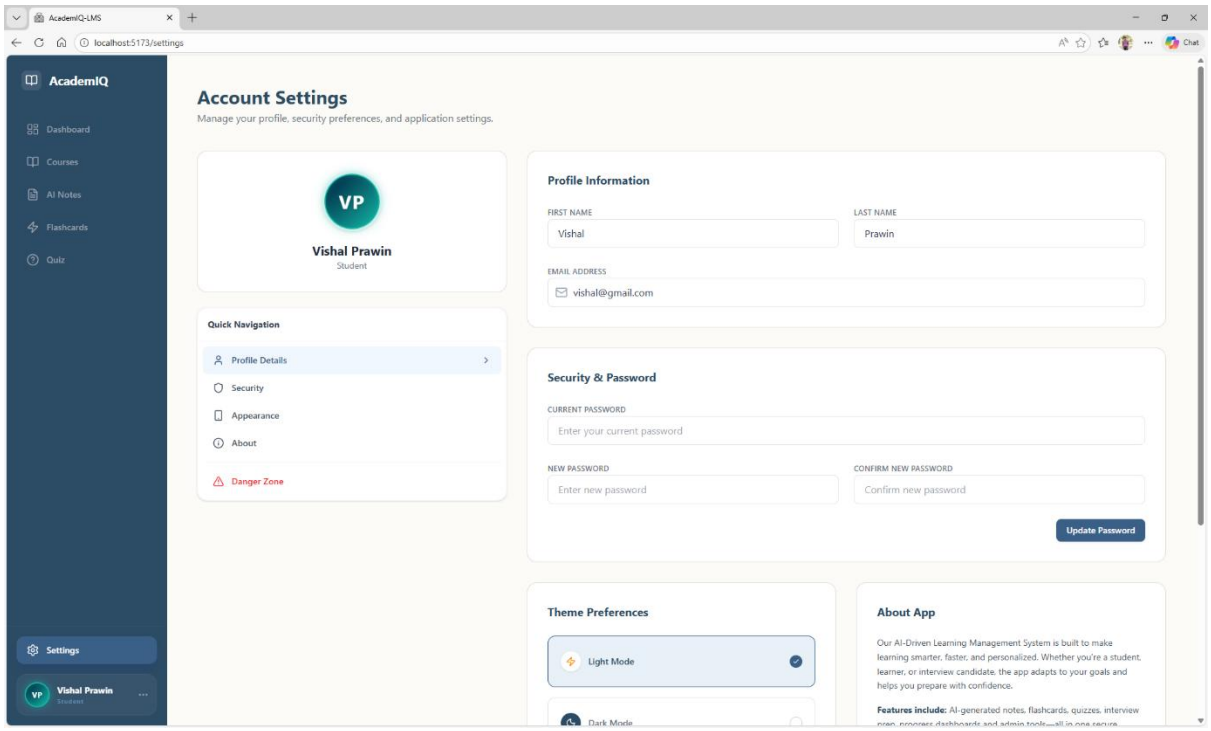

Figure. Settings Page

SAMPLE CODE:

```
"use client";
import { Button } from "@/components/ui/button";
import { supabase } from "@/services/supabaseClient";
import Image from "next/image";
import React from "react";


function Login() {

  const signInWithGoogle = async () => {
    const { error } = await supabase.auth.signInWithOAuth({
      provider: 'google',
    })
    if (error) {
      console.error('Error', error.message)
    }
  }

  return (
    <div className="flex flex-col items-center justify-center h-screen">
      <div className="flex flex-col items-center border rounded-2xl p-8">
        <Image
          src={"/logo.png"}
          alt="logo"
          width={400}
          height={100}
          className="w-[180px]"
        />
        <div className="flex items-center flex-col">
          <Image
            src={"/login.jpg"}
            alt="logo"
            width={600}
            height={400}
            className="w-[400px] h-[500px]"
          />
          <h2 className="text-2xl font-bold text-center mt-5">
            Welcome to AiCruit
          </h2>
          <p className="text-gray-500 text-center">
            Sign In With Google Authentication
          </p>
          <Button className="mt-7 w-full" onClick={signInWithGoogle}>
            {" "}
            Login With Google
          </Button>
        </div>
      </div>
```

```
    </div>
  );
}

export default Login;


import { QUESTION_PROMPT } from "@/services/Constants";
import { NextResponse } from "next/server";
import OpenAI from "openai";

export async function POST(req) {
  const { jobPosition, jobDescription, duration, type } = await req.json();

  const FINAL_PROMPT = QUESTION_PROMPT.replace("{{jobTitle}}", jobPosition)
    .replace("{{jobDescription}}", jobDescription)
    .replace("{{duration}}", duration)
    .replace("{{type}}", type);
  console.log(FINAL_PROMPT);

  try {
    const openai = new OpenAI({
      baseURL: "https://openrouter.ai/api/v1",
      apiKey: process.env.OPENROUTER_API_KEY,
    });

    const completion = await openai.chat.completions.create({
      model: "openai/gpt-4o-mini",
      messages: [
        {
          role: "user",
          content: FINAL_PROMPT,
        },
      ],
    });

    return NextResponse.json(completion.choices[0].message);
  } catch (e) {
    console.log(e);
    return NextResponse.json(e);
  }
}

import { Phone, Video } from "lucide-react";
import React from "react";
import Link from "next/link";
function CreateOptions() {
  return (
    <div className="grid grid-cols-2 gap-5">
      <Link
```

```
      href={"/dashboard/create-interview"}
      className="bg-white border border-gray-200 rounded-lg p-5 flex flex-col gap-2 cursor-
pointer"
    >
      <Video className="p-3 text-primary bg-blue-50 rounded-lg h-12 w-12" />
      <h2 className="font-bold">Create New Interview</h2>
      <p className="text-gray-500">
        Create AI Interviews and schedule then with Candidates
      </p>
    </Link>
    <div className="bg-white border border-gray-200 rounded-lg p-5 flex flex-col gap-2">
      <Phone className="p-3 text-primary bg-blue-50 rounded-lg h-12 w-12" />
      <h2 className="font-bold">Create Phone Screening Call</h2>
      <p className="text-gray-500">
        Schedule phone screening call with Candidates
      </p>
    </div>
  </div>
 );
}

export default CreateOptions;


  "use client";
import { Button } from "@/components/ui/button";
import { Camera, Video } from "lucide-react";
import React, { useState } from "react";

function LatestInterviewsList() {
 const [interviewList, setInterviewList] = useState([]);
 return (
  <div className="my-5">
    <h2 className="font-bold text-2xl">Previously Created Interviews</h2>
    {interviewList?.length == 0 && (
      <div className="p-5 flex flex-col gap-3 items-center mt-5">
        <Video className="h-10 w-10 text-primary" />
        <h2>You don't have any interview created!</h2>
        <Button>+ Create New Interview</Button>
      </div>
    )}
  </div>
 );
}

export default LatestInterviewsList;

"use client";
import { Progress } from "@/components/ui/progress";
import { ArrowLeft } from "lucide-react";
```

```
import { useRouter } from "next/navigation";
import React, { useState } from "react";
import FormContainer from "./_components/FormContainer";
import QuestionList from "./_components/QuestionList";
import { toast } from "sonner";
import InterviewLink from "./_components/InterviewLink";

function page() {
  const router = useRouter();
  const [step, setStep] = useState(1);
  const [formData, setFormData] = useState();
  const [interviewId, setInterviewId] = useState();
  const onHandleInputChange = (field, value) => {
    setFormData((prev) => ({
      ...prev,
      [field]: value,
    }));
    console.log("formData", formData);
  };

  const onGoToNext = () => {
    if (
      !formData?.jobPosition ||
      !formData?.jobDescription ||
      !formData?.duration ||
      !formData?.type
    ) {
      toast("Please enter all details!");
      return;
    }
    setStep(step + 1);
  };

  const onCreateLink = (interview_id) => {
    setInterviewId(interview_id);
    setStep(step + 1);
  };
  return (
    <div className="mt-5 px-10 md:px-24 lg:px-44 xl:px-56">
      <div className="flex gap-5 items-center">
        <ArrowLeft onClick={() => router.back()} className="cursor-pointer" />
        <h2 className="font-bold text-2xl">Create New Interview</h2>
      </div>
      <Progress value={step * 33.33} className="my-5" />
```

```
      {step === 1 ? (
        <FormContainer
          onHandleInputChange={onHandleInputChange}
          GoToNext={onGoToNext}
        />
      ) : step === 2 ? (
        <QuestionList
          formData={formData}
          onCreateLink={(interview_id) => onCreateLink(interview_id)}
        />
      ) : step === 3 ? (
        <InterviewLink interview_id={interviewId} formData={formData} />
      ) : null}
    </div>
  );
}

export default page;
```

SYNOPSIS

AI Recruiter is an AI-powered recruitment platform designed to modernize and streamline the hiring process through intelligent automation and a user-friendly interface. The system integrates modules such as interview scheduling, candidate evaluation, recruiter dashboards, and automated reporting into a single platform. It uses modern frontend technologies like Next.js and Tailwind CSS to deliver a responsive, fast, and seamless experience tailored specifically for recruiters.

The platform incorporates secure authentication through Supabase Google Auth, ensuring frictionless login and secure session management. Recruiters are provided with a dedicated dashboard that centralizes interview creation, candidate tracking, and evaluation reports, enabling structured workflows and controlled access. The backend, powered by Node.js, Express.js, and Supabase's managed PostgreSQL, efficiently handles candidate data while maintaining scalability, security, and smooth API communication across the system.

A key highlight of AI Recruiter is its integration with advanced AI services such as GPT-based models for dynamic interview question generation and Vapi for voice agent functionality and speech-to-text transcription. These intelligent features enable automated interviews, structured evaluation reports, and personalized candidate feedback. By combining a powerful technical stack with Supabase authentication and smart AI capabilities, AI Recruiter delivers a next-generation recruitment solution that is efficient, interactive, and adaptable to the evolving needs of modern hiring practices.

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